

9-bit SAR ADC Design

Mixed Signal VLSI Design

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Specifications

- Technology: 180 nm CMOS
- Resolution: 9-bit
- Sampling Rate: 20 MS/s
- Input Range: 0–0.8 V
- Analog Supply: 1.8 V
- Digital Supply: 1.2 V
- Architecture: SAR ADC
- DAC Type: Binary-weighted capacitive DAC

Working Principle

SAR ADC works using a **binary search algorithm**.

Sample → Compare → DAC Update → SAR Logic → Digital Output

- Input voltage is sampled and held.
- Comparator compares sampled input with DAC output.
- SAR logic updates bits from MSB to LSB.
- Final 9-bit digital output is generated.

Sample-and-Hold Circuit

Function:

- Samples analog input during sampling phase.
- Holds the voltage constant during conversion.

Design Features:

- Bootstrapped switch used for better linearity.
- Reduces ON-resistance variation.
- Minimizes charge injection and clock feedthrough.
- Provides accurate sampling for low-voltage operation.

Thermal Noise Calculation

Thermal noise requirement:

$$\frac{kT}{C} < \left(\frac{LSB}{2} \right)^2$$

LSB value:

$$LSB = \frac{V_{FS}}{2^N}$$

$$LSB = \frac{0.8}{2^9} = \frac{0.8}{512}$$

$$LSB = 1.56 \text{ mV}$$

$$C > \frac{1.38 \times 10^{-23} \times 300}{(0.78 \times 10^{-3})^2}$$

$$C > 6.8 \text{ fF}$$

Reason for Choosing 10 pF

Although the calculated minimum capacitance is only 6.8 fF, a practical value of 10 pF is selected.

- Reduces thermal noise.
- Improves capacitor matching.
- Gives better accuracy.
- Provides stable sampled voltage.
- Easier for practical circuit implementation.

Comparator

Function:

The comparator compares:

$$V_{IN} \text{ with } V_{DAC}$$

$$V_{IN} > V_{DAC} \Rightarrow \text{Output} = 1$$

$$V_{IN} < V_{DAC} \Rightarrow \text{Output} = 0$$

Design:

- High-speed dynamic comparator.
- Operates with 1.8 V analog supply.
- Differential operation improves noise immunity.

Comparator Design Considerations

Important design points:

- Low offset voltage.
- Reduced kickback noise.
- Fast regeneration.
- Stable operation at 20 MS/s.
- Accurate decision even for small voltage differences.

Function:

SAR logic performs bit-by-bit approximation from MSB to LSB.

- MSB is initially set to 1.
- DAC output is generated.
- Comparator result decides whether bit is kept or cleared.
- Process repeats for all 9 bits.

For 9-bit ADC:

9 comparison(sar) cycles are required

Digital Output Code

For an ideal ADC:

$$D = \left\lfloor \frac{V_{IN}}{V_{FS}} \times 2^N \right\rfloor$$

For this ADC:

$$D = \left\lfloor \frac{V_{IN}}{0.8} \times 512 \right\rfloor$$

Example:

$$V_{IN} = 0.4 \text{ V}$$

$$D = \left\lfloor \frac{0.4}{0.8} \times 512 \right\rfloor = 256$$

$$D = 100000000_2$$

Simulation Results

- ADC successfully converts analog input into 9-bit digital output.
- Comparator decisions correctly update SAR logic.
- DAC output converges toward sampled input voltage.
- Stable operation observed for input range 0–0.8 V.
- Resolution achieved: 9-bit.

Conclusion

- Designed a 9-bit SAR ADC using 180 nm CMOS technology.
- Implemented Sample-and-Hold, Comparator, SAR Logic, and Capacitive DAC.
- Achieved successful analog-to-digital conversion.
- Verified operation through simulation results.
- Suitable for low-power and moderate-speed mixed-signal applications.